

GATE DA Practice Set 2 — Covariance/Correlation/CLT, k-NN, Simple & Compound Interest

Coverage: Covariance, Correlation, Central Limit Theorem; k-NN Algorithm; Simple & Compound Interest.

Sourcing note: As with Set 1, labels are honest: **PYQ-PATTERN** = mirrors a well-known real GATE-style question structure/numbers but is not a verified exact paper question; **ORIGINAL** = newly written, GATE DA-level. No fabricated years or question numbers are used anywhere.

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1 1. Covariance, Correlation, CLT

Q1 MCQ PYQ-PATTERN EASY

The correlation coefficient $\rho(X,Y)$ always lies in the range:

- (A) $[0, 1]$
- (B) $[-1, 1]$
- (C) $(-\infty, \infty)$
- (D) $[-1, 0]$

ANSWER(B)

SOLUTION

By the Cauchy-Schwarz inequality, correlation is bounded between -1 and 1 .

Concept: Range of correlation coefficient

Q2 NAT PYQ-PATTERN MEDIUM

$\text{Cov}(X,Y)=6$, $\sigma_X=2$, $\sigma_Y=3$. Find the correlation coefficient $\rho(X,Y)$.

ANSWER1.0

SOLUTION

$$\rho = \text{Cov}(X,Y)/(\sigma_X \cdot \sigma_Y) = 6/(2 \times 3) = 1.$$

Concept: Correlation from covariance and standard deviations

Q3 NAT PYQ-PATTERN MEDIUM

$\text{Var}(X)=4$, $\text{Var}(Y)=9$, $\text{Cov}(X,Y)=2$. Find $\text{Var}(X+Y)$.

ANSWER17

SOLUTION

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y) = 4 + 9 + 4 = 17.$$

Concept: Variance of a sum

Q4 NAT PYQ-PATTERN MEDIUM

$\text{Var}(X)=4$, $\text{Var}(Y)=9$, $\text{Cov}(X,Y)=2$. Find $\text{Var}(X-Y)$.

ANSWER9

SOLUTION

$$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X,Y) = 4 + 9 - 4 = 9.$$

Concept: Variance of a difference

Q5 MCQ PYQ-PATTERN EASY

The Central Limit Theorem states that, for i.i.d. random variables with finite variance, as sample size $n \rightarrow \infty$:

- (A) The population distribution becomes normal
- (B) The sample mean's distribution approaches normal, regardless of the population's shape
- (C) The sample variance approaches 0
- (D) Every individual sample value becomes normal

ANSWER(B)

SOLUTION

CLT is about the sampling distribution of the mean approaching normality, not the population itself.

Concept: Statement of the Central Limit Theorem

Q6 NAT PYQ-PATTERN EASY

Population $\sigma=20$. Sample size $n=100$. Find the standard error of the sample mean.

ANSWER2**SOLUTION**

$$SE = \sigma/\sqrt{n} = 20/\sqrt{100} = 20/10 = 2.$$

Concept: Standard error formula

Q7 MSQ PYQ-PATTERN MEDIUM

Which of the following are TRUE about the correlation coefficient?

- (A) It is unit-less
- (B) $\text{corr}(X,Y) = \text{corr}(Y,X)$
- (C) $\rho = 1$ implies a perfect positive linear relationship
- (D) $\rho = 0$ implies X and Y are independent

ANSWER(A), (B), (C)**SOLUTION**

$\rho=0$ only rules out *linear* association; X and Y can still be dependent non-linearly, so (D) is false. (A), (B), (C) are standard properties.

Concept: Properties and common misconceptions of correlation

Q8 NAT PYQ-PATTERN MEDIUM

$\rho(X,Y)=0.8$, $\sigma_X=5$, $\sigma_Y=10$. Find $\text{Cov}(X,Y)$.

ANSWER40**SOLUTION**

$$\text{Cov}(X,Y) = \rho \cdot \sigma_X \cdot \sigma_Y = 0.8 \times 5 \times 10 = 40.$$

Concept: Covariance from correlation

Q9 NAT PYQ-PATTERN HARD

Population mean $\mu=50$, $\sigma=12$. A sample of size $n=36$ is drawn. Using the CLT, find $P(\text{sample mean} > 53)$ (round to 4 decimals).

ANSWER0.0668**SOLUTION**

$$SE = 12/\sqrt{36} = 2. \quad z = (53-50)/2 = 1.5. \quad P(\bar{X} > 53) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668.$$

Concept: CLT applied to a sampling-distribution probability

Q10 NAT ORIGINAL EASY

$\text{Var}(X)=16$, $\text{Var}(Y)=25$, $\text{Cov}(X,Y)=10$. Find the correlation coefficient $\rho(X,Y)$.

ANSWER0.5**SOLUTION**

$$\sigma_X=4, \sigma_Y=5. \quad \rho = 10/(4 \times 5) = 10/20 = 0.5.$$

Concept: Correlation calculation

Q11 MCQ ORIGINAL EASY

If X and Y are independent random variables, then:

- (A) $\text{Cov}(X, Y) = 1$
- (B) $\text{Cov}(X, Y) = 0$
- (C) $\text{Cov}(X, Y) = \text{Var}(X) + \text{Var}(Y)$
- (D) Cannot be determined

ANSWER(B)**SOLUTION**

Independence implies $E[XY] = E[X]E[Y]$, so $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$.

Concept: Independence implies zero covariance

Q12 NAT ORIGINAL MEDIUM

$E[X]=5$, $E[Y]=4$, $E[XY]=22$. Find $\text{Cov}(X, Y)$.

ANSWER2**SOLUTION**

$\text{Cov}(X, Y) = 22 - (5 \times 4) = 22 - 20 = 2$.

Concept: Covariance formula

Q13 NAT ORIGINAL MEDIUM

$\text{Var}(X)=4$, $\text{Var}(Y)=9$, $\text{Cov}(X, Y)=1$. Find $\text{Var}(2X - 3Y)$.

ANSWER85**SOLUTION**

$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab \cdot \text{Cov}(X, Y)$, with $a=2$, $b=-3$:
 $= 4 \times 4 + 9 \times 9 + 2(2)(-3)(1) = 16 + 81 - 12 = 85$.

Concept: Variance of a general linear combination

Q14 MCQ ORIGINAL MEDIUM

Which of the following is a necessary condition for the Central Limit Theorem to apply to the sample mean?

- (A) The population must be normally distributed
- (B) The population must have a finite variance
- (C) The sample size must be exactly 30
- (D) The population must be symmetric

ANSWER(B)**SOLUTION**

CLT requires finite population variance; it does not require the population itself to be normal, symmetric, or the sample size to be exactly 30 (30 is just a common rule-of-thumb, not a strict requirement).

Concept: Conditions under which CLT holds — conceptual trap

Q15 NAT ORIGINAL MEDIUM

Population $\sigma=8$. Sample size $n=64$. Find the standard error of the sample mean.

ANSWER1**SOLUTION**

$SE = 8/\sqrt{64} = 8/8 = 1$.

Concept: Standard error formula

Q16 NAT ORIGINAL MEDIUM

$\mu=100$, $\sigma=15$, $n=25$. Using the CLT, find $P(\text{sample mean} < 95)$ (round to 4 decimals).

ANSWER0.0478

SOLUTION
 $SE = 15/\sqrt{25} = 3$. $z = (95-100)/3 = -1.6667$. $P(\bar{X}<95) = \Phi(-1.6667) \approx 0.0478$.

Concept: CLT-based probability calculation

Q17 NAT ORIGINAL HARD

$\rho(X,Y)=0.6$, $\text{Var}(X)=9$, $\text{Var}(Y)=16$. Find $\text{Var}(X+Y)$.

ANSWER39.4

SOLUTION
 $\text{Cov}(X,Y) = 0.6 \times 3 \times 4 = 7.2$. $\text{Var}(X+Y) = 9+16+2(7.2) = 25+14.4 = 39.4$.

Concept: Combining correlation with variance to find Var of a sum

Q18 MCQ ORIGINAL HARD

Which formula correctly describes $\text{Cov}(aX+b, cY+d)$ for constants a, b, c, d ?

- (A) $ac \cdot \text{Cov}(X,Y)$
- (B) $ac \cdot \text{Cov}(X,Y) + bd$
- (C) $\text{Cov}(X,Y) + bd$
- (D) $a+c \cdot \text{Cov}(X,Y)$

ANSWER(A)

SOLUTION
Additive constants (b, d) don't affect covariance; multiplicative constants scale it: $\text{Cov}(aX+b, cY+d) = ac \cdot \text{Cov}(X,Y)$.

Concept: Covariance under affine transformation

Q19 NAT ORIGINAL HARD

A sample of size $n=50$ is drawn from a population with $\mu=200$, $\sigma=30$. Using the CLT, find $P(195 < \text{sample mean} < 205)$ (round to 4 decimals).

ANSWER0.7620

SOLUTION
 $SE = 30/\sqrt{50} \approx 4.2426$. $z_1 = (195-200)/4.2426 \approx -1.1785$, $z_2 = (205-200)/4.2426 \approx 1.1785$.
 $P = \Phi(1.1785) - \Phi(-1.1785) = 2\Phi(1.1785) - 1 \approx 2(0.8810) - 1 = 0.7620$.

Concept: CLT for an interval probability

Answer Key — Covariance, Correlation, CLT

1-1 | 2-B | 3-1.0 | 4-17 | 5-9 | 6-B | 7-2 | 8-ABC | 9-40 | 10-0.0668 | 11-0.5 | 12-B | 13-2 | 14-85 | 15-B | 16-1 | 17-0.0478 | 18-39.4 | 19-A | 20-0.7620

2 2. k-NN Algorithm

Q1 MCQ PYQ-PATTERN EASY

k-NN is best described as:

- (A) A parametric, eager learner
- (B) A non-parametric, lazy learner
- (C) A parametric, lazy learner
- (D) A non-parametric, eager learner

ANSWER(B)

SOLUTION

k-NN makes no assumption about the data's functional form (non-parametric) and defers computation to prediction time (lazy learner — no explicit training/model-fitting phase).

Concept: Fundamental characterization of k-NN

Q2 MCQ PYQ-PATTERN MEDIUM

As k increases in k-NN, the decision boundary generally becomes:

- (A) More jagged/complex
- (B) Smoother, and the model becomes less sensitive to noise
- (C) Undefined
- (D) Identical regardless of k

ANSWER(B)

SOLUTION

Larger k averages over more neighbors, smoothing the decision boundary and reducing variance (increasing bias).

Concept: Effect of k on bias-variance tradeoff

Q3 MSQ PYQ-PATTERN MEDIUM

For $k=1$ in k-NN (compared to a larger k), which of the following are TRUE?

- (A) The model has higher variance
- (B) The model has lower bias
- (C) The model is more sensitive to noisy/mislabeled points
- (D) The decision boundary is smoother

ANSWER(A), (B), (C)

SOLUTION

$k=1$ fits the training data very closely (low bias, high variance) and is highly sensitive to noise; its decision boundary is the most jagged, not smooth, so (D) is false.

Concept: $k=1$ special case — bias/variance and noise sensitivity

Q4 NAT PYQ-PATTERN EASY

Find the Manhattan distance between points (2,3) and (5,7).

ANSWER7

SOLUTION

$d = |5-2| + |7-3| = 3+4 = 7.$

Concept: Manhattan (L1) distance

Q5 MCQ PYQ-PATTERN MEDIUM

Training points: A(1,1)-Class1, B(2,2)-Class1, C(6,6)-Class2, D(7,7)-Class2, E(1,2)-Class1.
For test point T(2,1), using k=3 and Euclidean distance, predict the class.

- (A) Class1
- (B) Class2
- (C) Tie
- (D) Cannot be determined

ANSWER(A)

SOLUTION

Distances: A=1, B=1, E≈1.414, C≈6.4, D≈7.81. The 3 nearest are A, B, E — all Class1, so predicted class is Class1.

Concept: k-NN majority-vote classification, worked example

Q6 MSQ PYQ-PATTERN MEDIUM

Which of the following are TRUE about the k-NN algorithm?

- (A) It is computationally expensive at prediction (test) time
- (B) Feature scaling/normalization is typically important
- (C) It performs well in very high dimensions without any modification
- (D) It has no explicit training phase (it is a "lazy learner")

ANSWER(A), (B), (D)

SOLUTION

High-dimensional spaces suffer from the "curse of dimensionality," where distances become less meaningful — so (C) is false. (A), (B), (D) are standard, well-known facts about k-NN.

Concept: Practical characteristics and limitations of k-NN

Q7 NAT PYQ-PATTERN HARD

Using the Minkowski distance with p=3 between points (0,0) and (2,2), find the distance (round to 4 decimals).

ANSWER2.5198

SOLUTION

$d = (|2|^3 + |2|^3)^{1/3} = (8+8)^{1/3} = 16^{1/3} \approx 2.5198.$

Concept: Minkowski distance (generalized L-p norm)

Q8 MCQ PYQ-PATTERN EASY

For k-NN regression (as opposed to classification), the prediction for a test point is typically:

- (A) The mode of the k nearest neighbors' target values
- (B) The average of the k nearest neighbors' target values
- (C) The nearest single neighbor's value, always
- (D) The median of all training targets

ANSWER(B)

SOLUTION

In k-NN regression, the predicted value is the (often simple, sometimes weighted) average of the k nearest neighbors' target values.

Concept: k-NN for regression vs. classification

Q9 MCQ PYQ-PATTERN MEDIUM

As $k \rightarrow N$ (the total number of training points), the k-NN classifier's prediction for any test point tends toward:

- (A) The nearest neighbor's class only
- (B) The overall majority class in the training data (i.e., a constant prediction) — high bias, low variance
- (C) A random class
- (D) An undefined result

ANSWER(B)**SOLUTION**

When k equals the full training set size, every prediction is just a vote over the entire dataset, giving the same (majority-class) prediction everywhere — maximal bias, minimal variance.

Concept: Extreme case of k — bias-variance limit

Q10 NAT ORIGINAL EASY

Find the Euclidean distance between (0,0) and (3,4).

ANSWER5**SOLUTION**

$d = \sqrt{(9+16)} = \sqrt{25} = 5$.

Concept: Euclidean distance

Q11 MCQ ORIGINAL EASY

Compared to a larger k , $k=1$ in k-NN typically has:

- (A) Low variance, high bias
- (B) High variance, low bias
- (C) Low variance, low bias
- (D) High variance, high bias

ANSWER(B)**SOLUTION**

$k=1$ fits the training data extremely closely (low bias) but is very sensitive to which specific point is nearest (high variance).

Concept: Bias-variance tradeoff in k-NN

Q12 NAT ORIGINAL MEDIUM

Find the Manhattan distance between (1,5) and (4,1).

ANSWER7**SOLUTION**

$d = |4-1| + |1-5| = 3+4 = 7$.

Concept: Manhattan distance

Q13 MCQ ORIGINAL MEDIUM

Compared to $k=1$, choosing a larger value of k in k-NN typically:

- (A) Increases overfitting
- (B) Reduces overfitting (up to a point), by smoothing predictions
- (C) Has no effect on overfitting
- (D) Always leads to underfitting

ANSWER(B)**SOLUTION**

Larger k averages over more neighbors, reducing the model's sensitivity to individual noisy points and lowering variance — moderating overfitting (though too large a k can cause underfitting).

Concept: Effect of k on overfitting/underfitting

Q14

MCQ

ORIGINAL

MEDIUM

Training points: P1(1,1)-A, P2(2,3)-B, P3(3,3)-B, P4(6,5)-A, P5(7,8)-B.
For test point (3,2), using k=3 and Euclidean distance, predict the class.

- (A) Class A
- (B) Class B
- (C) Tie
- (D) Cannot be determined

ANSWER(B)

SOLUTION

Distances: $P1 \approx 2.236$, $P2 \approx 1.414$, $P3 = 1$, $P4 \approx 4.243$, $P5 \approx 7.211$. Nearest 3: P3(B), P2(B), P1(A) → majority is B (2 votes vs 1).

Concept: k-NN classification, worked multi-point example

Q15

MCQ

ORIGINAL

MEDIUM

In binary classification, k is often chosen to be an odd number primarily to:

- (A) Reduce computation time
- (B) Avoid ties in the majority vote
- (C) Guarantee higher accuracy
- (D) Satisfy a mathematical requirement of Euclidean distance

ANSWER(B)

SOLUTION

An odd k avoids the possibility of an exact tie between the two classes during the majority vote.

Concept: Practical rule of thumb for choosing k

Q16

MSQ

ORIGINAL

MEDIUM

Regarding feature scaling in k-NN, which of the following are TRUE?

- (A) It is necessary because distance metrics are sensitive to the scale of each feature
- (B) Without scaling, features with larger numeric ranges tend to dominate the distance calculation
- (C) Scaling is guaranteed to change the predicted class for every test point
- (D) Categorical variables typically need encoding (e.g., one-hot) before applying Euclidean distance

ANSWER(A), (B), (D)

SOLUTION

Scaling generally matters but doesn't necessarily change every prediction — (C) overstates its effect and is false. (A), (B), (D) are standard, correct practices.

Concept: Preprocessing requirements for k-NN

Q17

NAT

ORIGINAL

HARD

In weighted k-NN, weight = 1/distance. Three nearest neighbors have distances 1, 2, 4 with classes A, A, B respectively. Find the total weighted vote for class A.

ANSWER1.5

SOLUTION

$\text{Weight}_1 = 1/1 = 1$ (class A). $\text{Weight}_2 = 1/2 = 0.5$ (class A). Total for A = $1 + 0.5 = 1.5$.

Concept: Weighted k-NN vote calculation

Q18

MCQ

ORIGINAL

HARD

For naive (brute-force) k-NN, the time complexity of classifying one test point among n training points with d dimensions is:

- (A) O(d)
- (B) O(n)
- (C) O(nd)
- (D) O(n²d)

ANSWER(C)

SOLUTION
Computing the distance to each of the n training points takes O(d) time each, giving a total of O(nd) for one test point (before considering the cost of finding the k smallest).

Concept: Time complexity of brute-force k-NN

Q19

NAT

ORIGINAL

HARD

Using the Minkowski distance with p=4 between points (1,1) and (3,4), find the distance (round to 4 decimals).

ANSWER3.1385

SOLUTION
 $d = (|2|^4 + |3|^4)^{1/4} = (16 + 81)^{1/4} = 97^{0.25} \approx 3.1385$.

Concept: Minkowski distance with higher-order p

Answer Key — k-NN Algorithm

1-5 | 2-B | 3-B | 4-ABC | 5-7 | 6-A | 7-ABD | 8-2.5198 | 9-B | 10-B | 11-5 | 12-B | 13-7 | 14-B | 15-B | 16-B | 17-ABD | 18-1.5 | 19-C | 20-3.1385

3. Simple & Compound Interest

Q1 NAT PYQ-PATTERN EASY

Find the Compound Interest on P=Rs. 2000 at R=10% per annum for T=2 years (compounded annually).

ANSWER420

SOLUTION

$$A = P(1+R/100)^T = 2000 \times (1.1)^2 = 2000 \times 1.21 = 2420. \text{ CI} = A - P = 2420 - 2000 = 420.$$

Concept: Compound interest formula

Q2 NAT PYQ-PATTERN MEDIUM

A sum of Rs. 1000 earns Simple Interest of Rs. 300 in 5 years. Find the rate of interest (%).

ANSWER6

SOLUTION

$$R = (SI \times 100) / (P \times T) = (300 \times 100) / (1000 \times 5) = 30000 / 5000 = 6\%.$$

Concept: Solving for rate given SI

Q3 NAT PYQ-PATTERN MEDIUM

A sum of Rs. 1500 earns Simple Interest of Rs. 450 at 10% per annum. Find the time period (in years).

ANSWER3

SOLUTION

$$T = (SI \times 100) / (P \times R) = (450 \times 100) / (1500 \times 10) = 45000 / 15000 = 3 \text{ years}.$$

Concept: Solving for time given SI

Q4 NAT PYQ-PATTERN MEDIUM

Find the difference between CI and SI on Rs. 8000 at 5% per annum for 2 years.

ANSWER20

SOLUTION

$$\text{For 2 years, CI} - \text{SI} = P(R/100)^2 = 8000 \times (0.05)^2 = 8000 \times 0.0025 = 20.$$

Concept: CI-SI difference shortcut formula (2-year case)

Q5 NAT PYQ-PATTERN MEDIUM

Find the amount on Rs. 1000 at 10% per annum for 1 year, compounded half-yearly.

ANSWER1102.5

SOLUTION

$$\text{Half-yearly rate} = 5\%, \text{ number of periods} = 2. A = 1000 \times (1.05)^2 = 1000 \times 1.1025 = 1102.5.$$

Concept: Compounding at sub-annual periods

Q6 NAT PYQ-PATTERN HARD

Find the Compound Interest on Rs. 10,000 at 8% per annum for 1 year, compounded quarterly (round to 2 decimals).

ANSWER824.32

SOLUTION

$$\text{Quarterly rate} = 2\%, \text{ periods} = 4. A = 10000 \times (1.02)^4 = 10000 \times 1.082432 = 10824.32. \text{ CI} = 824.32.$$

Concept: Quarterly compounding

Q7 **NAT** **PYQ-PATTERN** **MEDIUM**

The Compound Interest on a sum for 2 years at 10% per annum is Rs. 420. Find the principal.

ANSWER2000

SOLUTION

$$A = P(1.1)^2 = 1.21P. CI = 0.21P = 420 \implies P = 420/0.21 = 2000.$$

Concept: Back-solving for principal from CI

Q8 **NAT** **PYQ-PATTERN** **MEDIUM**

At what rate of Simple Interest will a sum become 1.5 times itself (i.e., SI = half the principal) in 10 years?

ANSWER5

SOLUTION

$$SI = P/2. \text{ Using } SI = PRT/100: P/2 = (P \times R \times 10)/100 \implies R = 100/(2 \times 10) = 5\%.$$

Concept: SI applied to a "sum grows to a multiple" problem

Q9 **MCQ** **PYQ-PATTERN** **EASY**

For the same principal, rate, and a time period greater than 1 year, how does Compound Interest compare to Simple Interest?

- (A) CI is always less than SI
- (B) CI is always greater than SI (for $T > 1$, $R > 0$)
- (C) CI always equals SI
- (D) The comparison depends only on the principal

ANSWER(B)

SOLUTION

Since interest compounds on previously earned interest as well, $CI > SI$ whenever $T > 1$ and $R > 0$ (they're equal exactly at $T=1$).

Concept: CI vs. SI conceptual comparison

Q10 **NAT** **ORIGINAL** **EASY**

Find the Simple Interest on Rs. 3000 at 6% per annum for 4 years.

ANSWER720

SOLUTION

$$SI = (3000 \times 6 \times 4)/100 = 720.$$

Concept: Simple interest formula

Q11 **NAT** **ORIGINAL** **EASY**

Find the Compound Interest on Rs. 1000 at 10% per annum for 1 year (compounded annually).

ANSWER100

SOLUTION

$$A = 1000 \times 1.1 = 1100. CI = 100. \text{ (Note: for } T=1, CI \text{ equals } SI.)$$

Concept: CI for a single compounding period

Q12 **NAT** **ORIGINAL** **MEDIUM**

A sum earns Simple Interest of Rs. 600 in 4 years at 5% per annum. Find the principal.

ANSWER3000

SOLUTION

$$P = (SI \times 100)/(R \times T) = (600 \times 100)/(5 \times 4) = 60000/20 = 3000.$$

Concept: Solving for principal given SI

Q13 NAT ORIGINAL MEDIUM

Find the Compound Interest on Rs. 5000 at 12% per annum for 2 years (compounded annually).

ANSWER1272**SOLUTION**

$$A = 5000 \times (1.12)^2 = 5000 \times 1.2544 = 6272. \text{ CI} = 6272 - 5000 = 1272.$$

Concept: Two-year compound interest

Q14 NAT ORIGINAL MEDIUM

Find the difference between CI and SI on Rs. 5000 at 10% per annum for 2 years.

ANSWER50**SOLUTION**

$$\text{SI} = (5000 \times 10 \times 2) / 100 = 1000. \text{ CI: } A = 5000 \times 1.21 = 6050, \text{ CI} = 1050. \text{ Difference} = 1050 - 1000 = 50 \text{ (matches shortcut } P(R/100)^2 = 5000 \times 0.01 = 50).$$

Concept: CI-SI difference — verifying via both direct calculation and formula

Q15 NAT ORIGINAL MEDIUM

At what rate of Simple Interest will a sum double itself in 8 years?

ANSWER12.5**SOLUTION**

$$\text{Doubling means SI} = P. P = (P \times R \times 8) / 100 \implies R = 100 / 8 = 12.5\%.$$

Concept: SI rate for doubling in given time

Q16 MCQ ORIGINAL MEDIUM

In how many years will Rs. 1000 become Rs. 1331 at 10% per annum compound interest (compounded annually)?

- (A) 2 years
- (B) 3 years
- (C) 4 years
- (D) 5 years

ANSWER(B)**SOLUTION**

$$1000 \times (1.1)^n = 1331 \implies (1.1)^n = 1.331. \text{ Since } (1.1)^3 = 1.331 \text{ exactly, } n = 3.$$

Concept: Solving for time in compound interest

Q17 NAT ORIGINAL HARD

A sum, invested at compound interest (compounded annually) at 25% per annum, amounts to Rs. 15,625 after 3 years. Find the principal.

ANSWER8000**SOLUTION**

$$A = P(1.25)^3 = 15625 \implies P = 15625 / 1.953125 = 8000.$$

Concept: Back-solving principal over multiple compounding periods

Q18 NAT ORIGINAL HARD

Find the Compound Interest on Rs. 10,000 at 5% per annum for 3 years (compounded annually).

ANSWER1576.25**SOLUTION**

$$A = 10000 \times (1.05)^3 = 10000 \times 1.157625 = 11576.25. \text{ CI} = 11576.25 - 10000 = 1576.25.$$

Concept: Three-year compound interest calculation

Q19 NAT ORIGINAL HARD

A sum invested at compound interest (compounded annually) amounts to Rs. 2420 after 2 years and Rs. 2662 after 3 years. Find the rate of interest (%).

ANSWER10

SOLUTION

The ratio of amounts over one additional year gives $(1+R/100)$: $2662/2420 = 1.1 \implies R = 10\%$.

Concept: Extracting the rate from two consecutive-year amounts

Answer Key — Simple & Compound Interest

1-1200 | 2-420 | 3-6 | 4-3 | 5-20 | 6-1102.5 | 7-824.32 | 8-2000 | 9-5 | 10-B | 11-720 | 12-100 | 13-3000 | 14-1272 | 15-50 | 16-12.5 | 17-B | 18-8000 | 19-1576.25 | 20-10

COMPLETE ANSWER KEY — ALL 60 QUESTIONS

Covariance, Correlation, CLT: 1-1, 2-B, 3-1.0, 4-17, 5-9, 6-B, 7-2, 8-ABC, 9-40, 10-0.0668, 11-0.5, 12-B, 13-2, 14-85, 15-B, 16-1, 17-0.0478, 18-39.4, 19-A, 20-0.7620

k-NN Algorithm: 1-5, 2-B, 3-B, 4-ABC, 5-7, 6-A, 7-ABD, 8-2.5198, 9-B, 10-B, 11-5, 12-B, 13-7, 14-B, 15-B, 16-B, 17-ABD, 18-1.5, 19-C, 20-3.1385

Simple & Compound Interest: 1-1200, 2-420, 3-6, 4-3, 5-20, 6-1102.5, 7-824.32, 8-2000, 9-5, 10-B, 11-720, 12-100, 13-3000, 14-1272, 15-50, 16-12.5, 17-B, 18-8000, 19-1576.25, 20-10